

Chapter 2: Steady-State Converter Analysis

Erickson & Maksimović — Fundamentals of Power Electronics

Study Notes By Catastrophy_ame

2.1 Buck Converter and Small-Ripple Approximation

Buck Converter Topology

The switch controls whether energy is transferred to the load. An LC low-pass filter converts the switching waveform into a dc output.

- Switch at position 1 (on, duration DT_s): $v_s(t) = V_g$
- Switch at position 2 (off, duration $D'T_s$): $v_s(t) = 0$
- Switching frequency $f_s = 1/T_s$, typically 1 kHz – 1 MHz

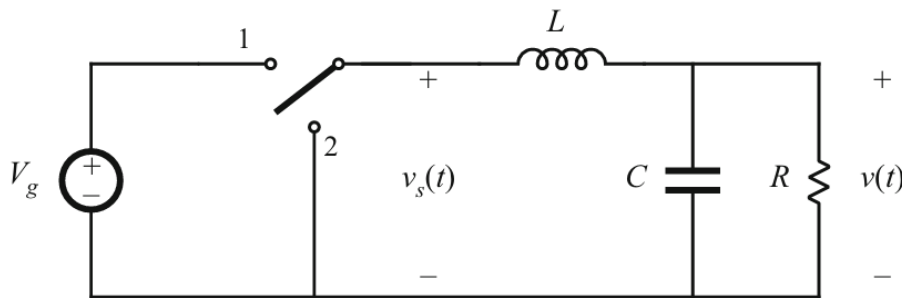


Figure 1: Buck Topology

Small-Ripple Approximation

The LC filter cannot perfectly remove switching harmonics. The actual output is:

$$v(t) = V + v_{\text{ripple}}(t)$$

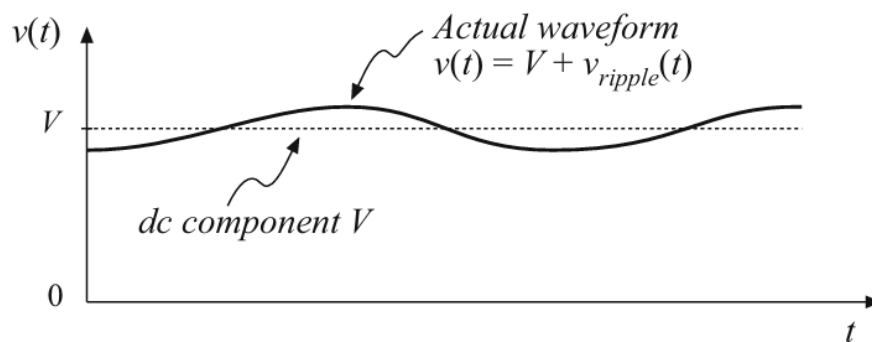


Figure 2: Output voltage waveform $v(t)$, consisting of dc V and switching ripple $v_{\text{ripple}}(t)$

In a well-designed converter the ripple is small (e.g., $< 1\%$ for a 3.3 V supply):

$$\|v_{\text{ripple}}\| \ll V \quad \Rightarrow \quad \boxed{v(t) \approx V}$$

This is the small-ripple (linear-ripple) approximation. It replaces exponential/damped-sinusoidal waveforms with simple linear segments, greatly simplifying analysis.

An exact periodic steady-state solution could in principle be obtained via Laplace transforms applied to each sub-circuit, followed by inverse transformation and boundary-condition matching.

Method	Advantage	Disadvantage
Exact (Laplace)	Theoretically complete	Intractable; results hard to interpret
Small-ripple approx.	Simple; clean closed-form	Ripple shape not captured

Valid for (continuous variables): inductor current $i_L(t)$; capacitor voltage $v_C(t)$.
Not valid for (discontinuous): switch voltage, switch current, inductor voltage.

Inductor Current Waveform (Buck)

Applying the approximation $v(t) \approx V$:

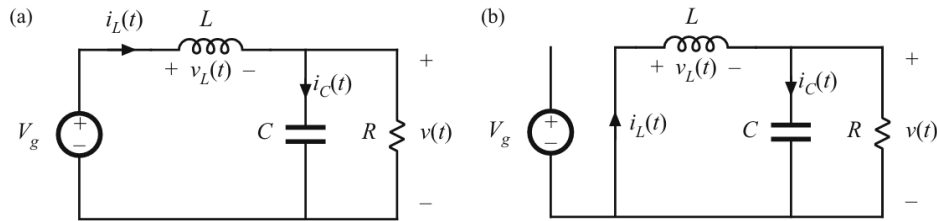


Figure 3: Buck Switching Stage

Interval	Inductor voltage	Current slope
Switch ON ($0 < t < DT_s$)	$v_L = V_g - V$	$\frac{di_L}{dt} = \frac{V_g - V}{L} > 0$
Switch OFF ($DT_s < t < T_s$)	$v_L = -V$	$\frac{di_L}{dt} = -\frac{V}{L} < 0$

Buck Inductor Selection

The ripple expression can be directly inverted to find the inductance value required for a specified ripple.

From the rising-slope relation:

$$2\Delta i_L = \frac{V_g - V}{L} \cdot DT_s$$

Solving for L :

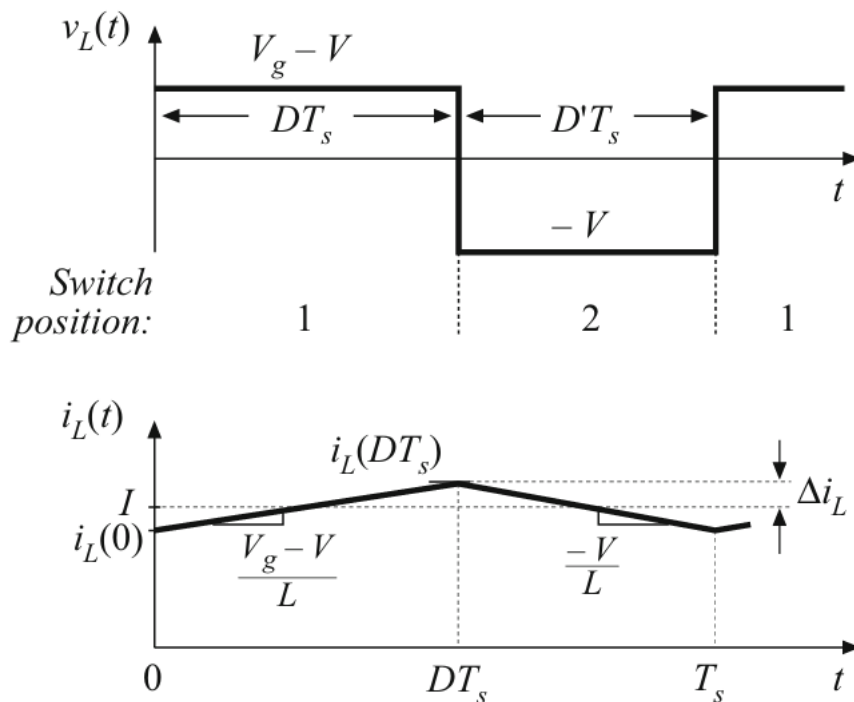


Figure 4: Steady-state inductor waveform

$$L = \frac{V_g - V}{2 \Delta i_L} DT_s$$

Design flow: specify the allowable peak ripple Δi_L (typically 10–20% of full-load current), then substitute known V_g , V , D , T_s to obtain the required inductance directly.

Why use the rising slope, not the falling slope?

Both intervals give identical results. The rising-slope form is chosen by convention, it contains $(V_g - V)$ and D , making substitution of design parameters more natural.

Start-Up Transient

We now examine what happens when the converter is first energised.

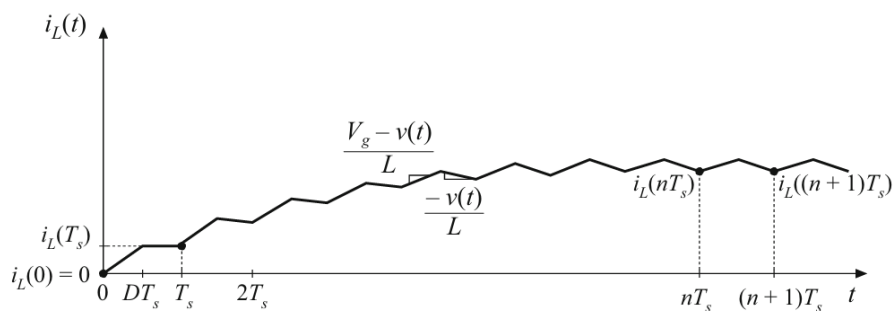


Figure 5: Inductor current waveform during converter turn-on transient

initial instant ($v = 0$, $i_L(0) = 0$):

- Interval-1 slope: $\frac{V_g - v}{L} \approx \frac{V_g}{L}$ (**steep**)
- Interval-2 slope: $-\frac{v}{L} \approx 0$ (**nearly zero**)

Result: net increase in i_L each cycle; output capacitor charges; v rises gradually.

Change	Effect
$v \uparrow$	Interval-1 slope $\frac{V_g - v}{L}$ decreases (slower rise)
$v \uparrow$	Interval-2 slope $-\frac{v}{L}$ becomes more negative (faster fall)

Steady state is reached when rise = fall over one period:

$$i_L(nT_s) = i_L((n+1)T_s)$$

Physical insight: volt-second balance is not an imposed constraint — it is the natural equilibrium to which the circuit converges. The system finds its own balance point.

2.2 Volt-Second Balance, Charge Balance, and Ripple

Inductor Volt-Second Balance

From $v_L(t) = L \frac{di_L}{dt}$, integrating over one period:

$$i_L(T_s) - i_L(0) = \frac{1}{L} \int_0^{T_s} v_L(t) dt$$

In steady state $i_L(T_s) = i_L(0)$, therefore:

$$0 = \frac{1}{T_s} \int_0^{T_s} v_L(t) dt = \langle v_L \rangle$$

The average (dc component) of inductor voltage must be zero in equilibrium.

Applying to Buck:

$$\langle v_L \rangle = D(V_g - V) + D'(-V) = DV_g - V = 0 \quad \Rightarrow \quad V = DV_g \checkmark$$

Capacitor Charge (Amp-Second) Balance

Identical argument applied to $i_C(t) = C \frac{dv_C}{dt}$:

$$0 = \frac{1}{T_s} \int_0^{T_s} i_C(t) dt = \langle i_C \rangle$$

The average capacitor current must be zero in steady state.

Inductor Current Ripple (Buck)

From the rising slope and interval length:

$$2\Delta i_L = \frac{V_g - V}{L} \cdot DT_s \quad \Rightarrow \quad \boxed{\Delta i_L = \frac{V_g - V}{2L} DT_s}$$

Solving for L to achieve a desired ripple:

$$L = \frac{V_g - V}{2\Delta i_L} DT_s$$

2.3 Boost Converter

Topology

Inductor and switch positions are swapped relative to buck. Output voltage is always greater than or equal to input.

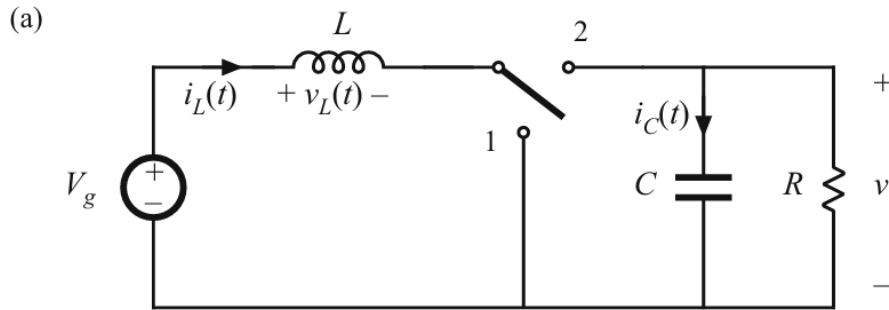


Figure 6: Boost Topology

Steady-State Analysis

Interval	Inductor voltage	Diode
Switch ON (DT_s)	$v_L = V_g$	reverse biased
Switch OFF ($D'T_s$)	$v_L = V_g - V$	forward biased

Volt-second balance:

$$\langle v_L \rangle = D V_g + D'(V_g - V) = V_g - D'V = 0$$

$$\boxed{V = \frac{V_g}{D'} = \frac{V_g}{1 - D}, \quad V \geq V_g}$$

DC Input Current

By charge balance on the output capacitor and power conservation:

$$\langle i_L \rangle = I = \frac{V}{D'R} = \frac{I_{out}}{D'}$$

Since $V > V_g$, we have $I > I_{out}$ — consistent with power conservation.

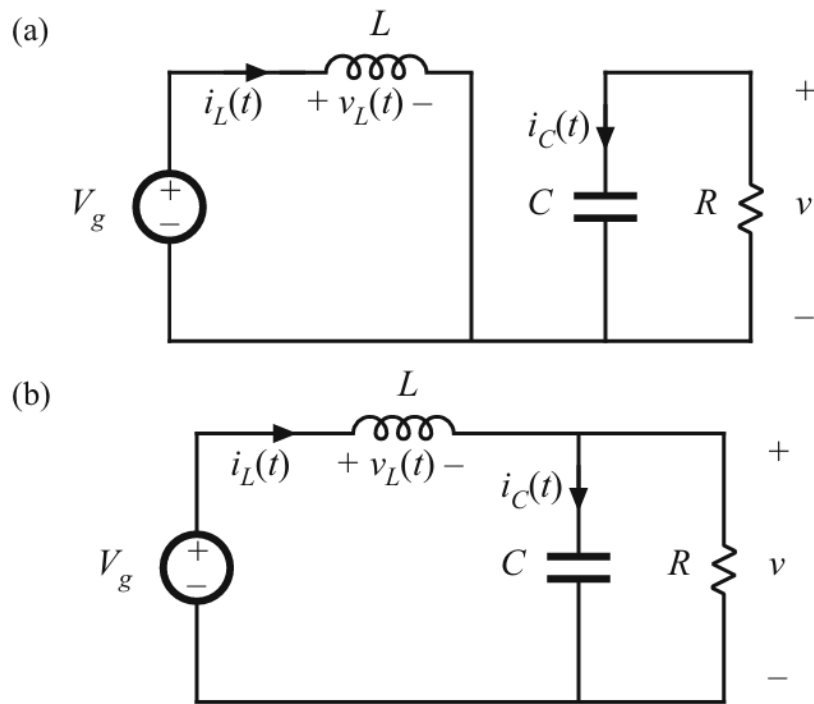


Figure 7: Boost Switching Stage

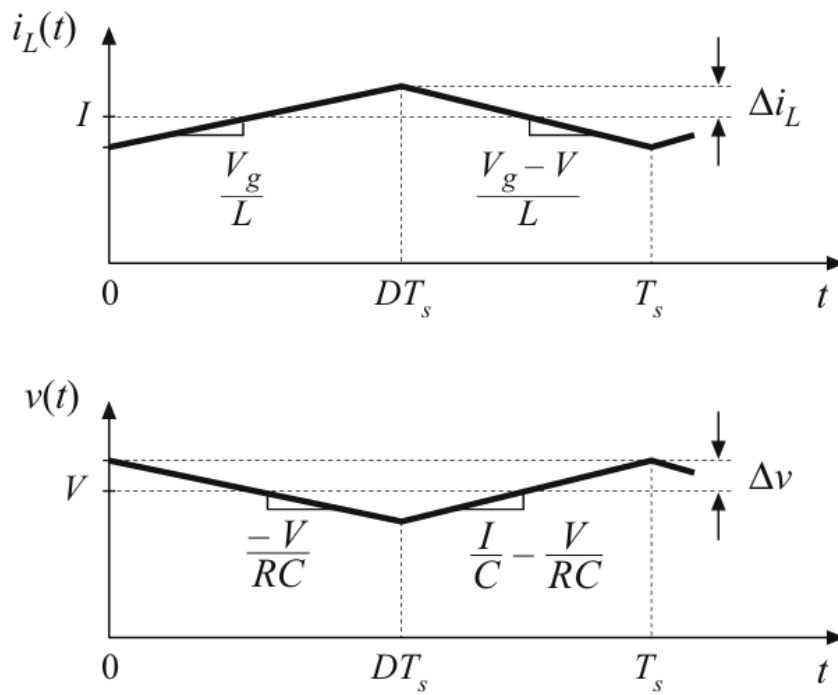


Figure 8: Boost inductor waveform

Current Ripple

$$\Delta i_L = \frac{V_g}{2L} DT_s$$

High Duty-Cycle Limitation

As $D \rightarrow 1$: $I_L \rightarrow \infty$. Conduction losses ($I^2 R_{\text{loss}}$) grow rapidly, causing efficiency to drop sharply. **This is the fundamental practical limit of the boost converter.**

The boost capacitor current is pulsating, so the ripple is found directly from the slope–interval product.

Interval	Capacitor current i_C	Voltage slope $\frac{dv_C}{dt}$
Switch ON (DT_s)	$-V/R$ (load discharge only)	$-\frac{V}{RC}$
Switch OFF ($D'T_s$)	$I - V/R$ (inductor supplies load)	$\frac{I}{C} - \frac{V}{RC}$

Voltage change during the on-interval equals slope $\times$ duration:

$$-2\Delta v = \frac{-V}{RC} \cdot DT_s$$

$$\Delta v = \frac{V}{2RC} DT_s$$

Use this expression to select C for a specified peak ripple Δv .

2.4 Ćuk Converter

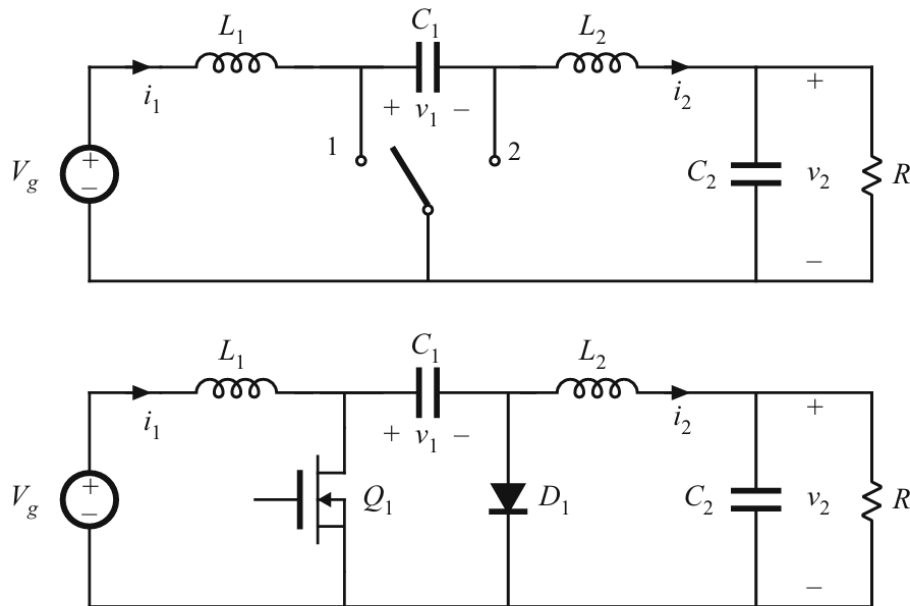
Topology and Key Feature

Both input and output have series inductors, giving continuous (non-pulsating) current at both terminals. Energy is transferred via the intermediate capacitor C_1 .

	Buck-Boost	Ćuk
Conversion ratio	$-D/D'$	$-D/D'$
Output polarity	inverted	inverted
Input current	pulsating	continuous
Output current	pulsating	continuous
Energy storage	1 inductor	2 inductors + 1 capacitor

Steady-State Analysis

	Switch ON (DT_s)	Switch OFF ($D'T_s$)
v_{L1}	V_g	$V_g - V_{C1}$
v_{L2}	$-V_{C1} - V$	$-V$



Cuk converter: (a) with ideal switch, (b) practical realization using MOSFET and diode

Figure 9: Cuk Topology

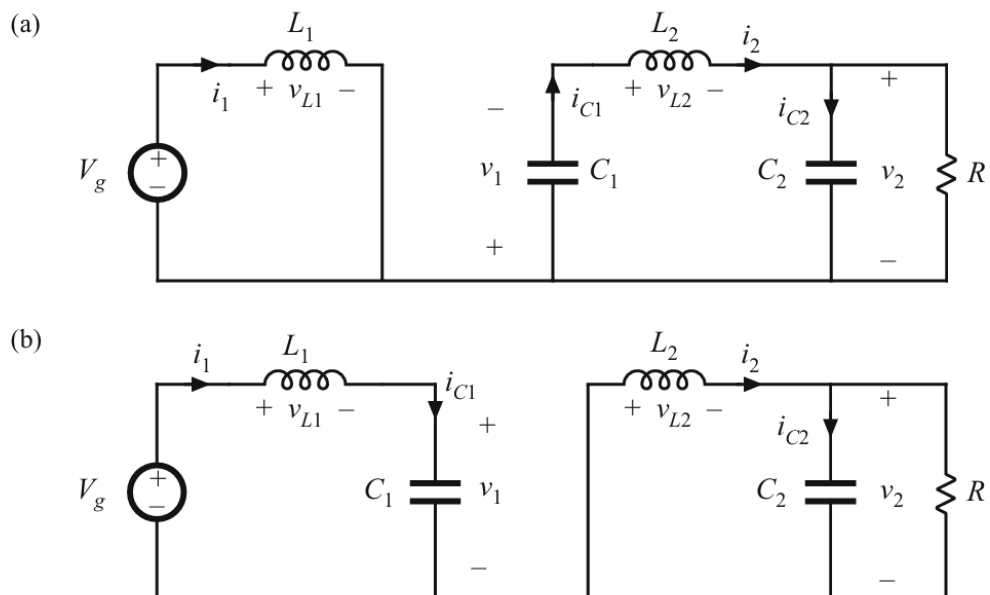


Figure 10: Cuk Switching Stage

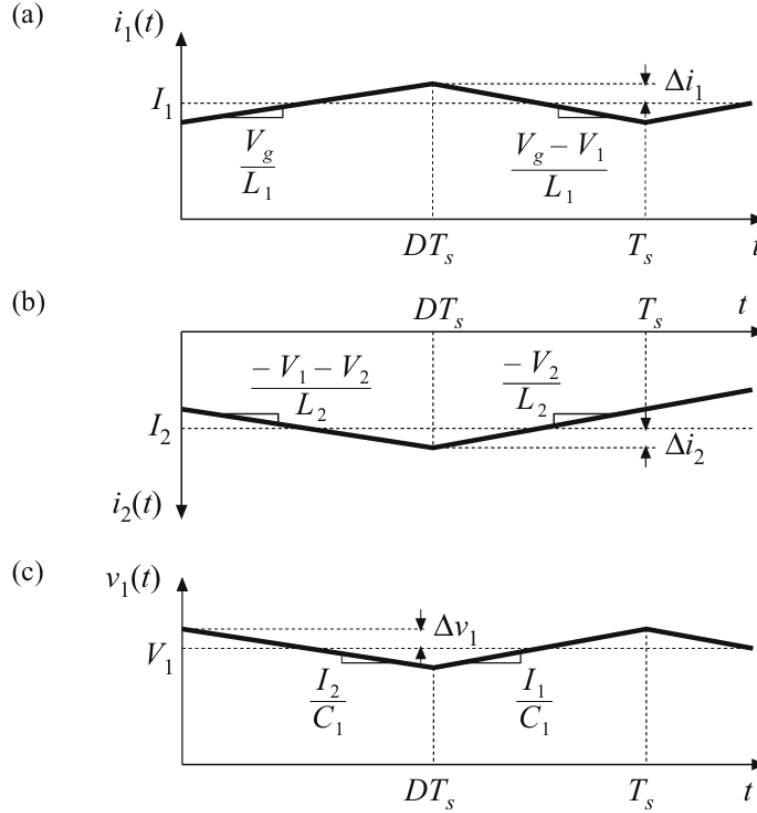


Figure 11: Cuk Converter Waveform

Volt-second balance on L_1 : $DV_g + D'(V_g - V_{C1}) = 0 \Rightarrow V_{C1} = \frac{V_g}{D'}$

Volt-second balance on L_2 : $D(-V_{C1} - V) + D'(-V) = 0 \Rightarrow V = -\frac{D}{D'}V_g$

$$V = -\frac{D}{D'} V_g, \quad V_{C1} = V_g + |V| = \frac{V_g}{D'}$$

Output polarity is inverted. $|V_{C1}|$ is the highest voltage in the circuit.

Current Ripples

$$\Delta i_{L1} = \frac{V_g}{2L_1} DT_s, \quad \Delta i_{L2} = \frac{|V|}{2L_2} D'T_s, \quad \Delta v_{C1} = \frac{I_2}{2C_1} DT_s$$

Output Capacitor Ripple — Special Case

The current $i_{C2}(t)$ is continuous (non-pulsating). Direct small-ripple approximation gives $\Delta v_2 = 0$ — incorrect. The only ac source driving C_2 is the ripple of i_{L2} .

Do not apply the small-ripple approximation to i_{L2} when computing output ripple. Use the method of the next section instead.

2.5 Output Capacitor Voltage Ripple

Method: Charge Integral of Triangular Current

When the capacitor current $i_C(t)$ is a triangular wave with peak Δi_L , the voltage ripple is found from the charge deposited between zero-crossings.

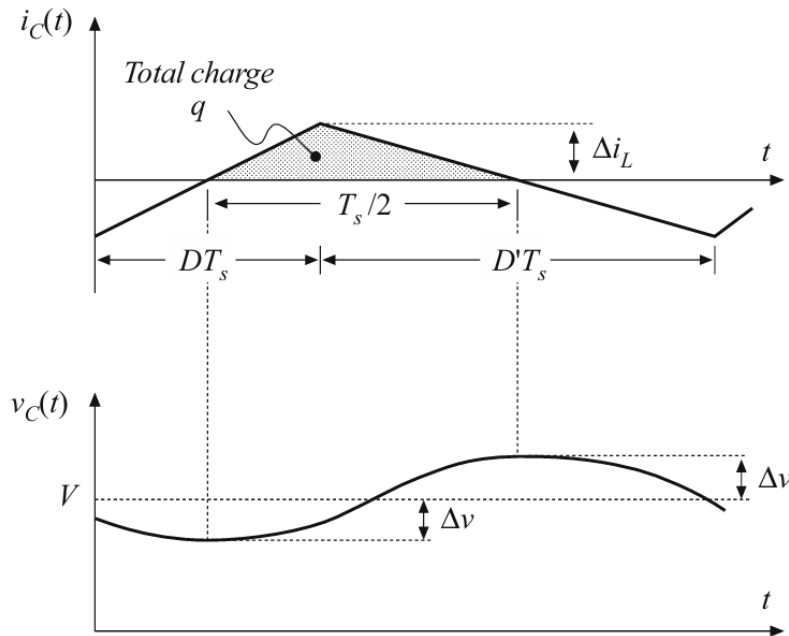


Figure 12: Output capacitor voltage and current waveforms

By symmetry, the zero crossings of $i_C(t)$ occur at the midpoints of each subinterval. The shaded triangle has:

- Height: Δi_L
- Base: $T_s/2$

Total charge in positive half-cycle:

$$q = \frac{1}{2} \Delta i_L \cdot \frac{T_s}{2}$$

Using $q = C(2\Delta v)$:

$$\Delta v = \frac{\Delta i_L T_s}{8C}$$

To reduce ripple: increase C , increase L (reduces Δi_L), or increase f_s .

Continuous Inductor Voltage Case

When $v_L(t)$ is continuous (as in the Ćuk output side), a dual argument applies using flux linkage $\lambda = Li$. The result has the same form with $\Delta v \leftrightarrow \Delta i$ and $C \leftrightarrow L$.

2.6 Summary of Key Principles

1. **DC component** of a waveform = average over one switching period.
2. **Small-ripple approximation:** $v(t) \approx V$, $i_L(t) \approx I_L$ — valid only for continuous variables (inductor current, capacitor voltage).
3. **Inductor volt-second balance:** $\langle v_L \rangle = 0$ in steady state — used to find dc voltages.
4. **Capacitor charge balance:** $\langle i_C \rangle = 0$ in steady state — used to find dc currents.
5. **Ripple** is computed from slopes and intervals; capacitor ripple uses the charge-triangle method.